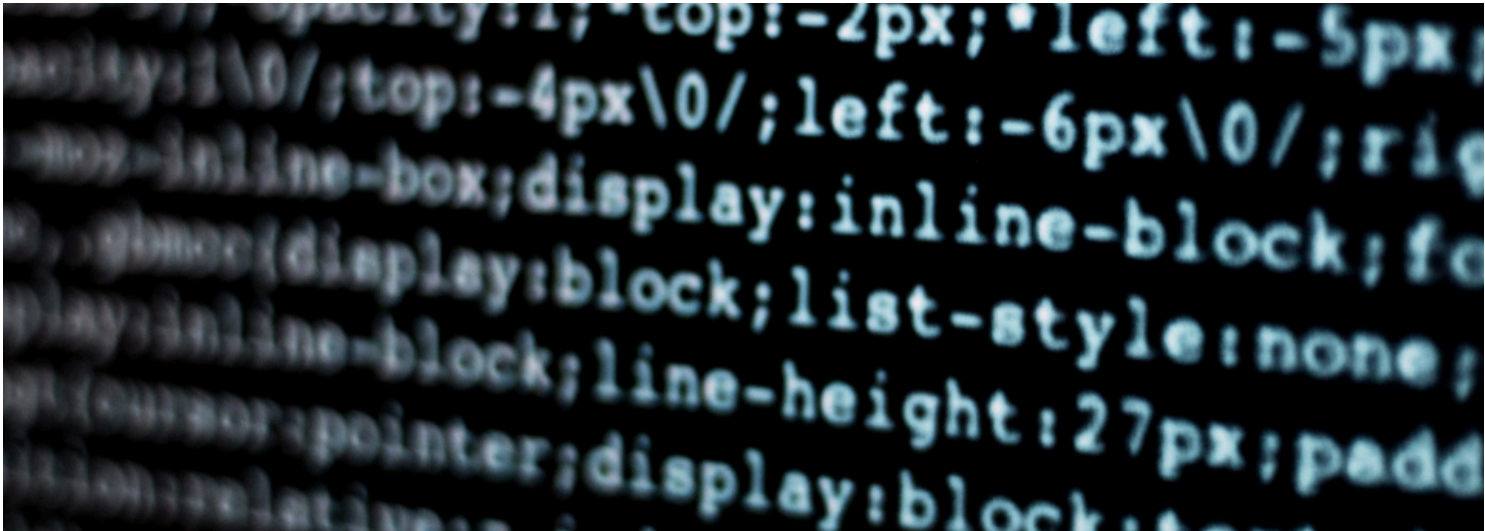




GS-B-WERC_DATA_MANAGEMENT@USGS.GOV

Software Releases



WERC's Main SharePoint



GitLab Ticket

Contacts:



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Click the images above to check out our guides.

Accessing GitLab

Creating a GitLab Account (USGS)

- USGS employee uses single sign on to initiate log on of USGS GitLab (https://code.usgs.gov/users/sign_in).
- USGS employee emails the WERC Data Management group (gs-b-werc_data_management@usgs.gov) and requests membership to the WERC group.
- Employee will receive an email stating they have been added to WERC GitLab group.

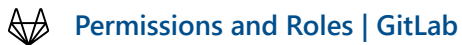
Accessing GitLab (External DOI Collaborators)

- USGS employee *sponsor* [submits a ticket](#) through USGS Software Management.
- When you submit a New Ticket, under Description select *External Collaborator*. Fill out the form that appears below.
- Upon submitting this request, an administrator will create the account for the external collaborator. All external accounts require two factor authentication. It is up to the user to enable/configure this on their account, and they have 48 hours to do so. Following this time period, the account will be automatically disabled, and additional administrative action will be required.
- The Collaborator will receive an email with additional instructions

after your ticket has been processed.

- The USGS *sponsor* should then email the WERC Data Management group (gs-b-werc_data_management@usgs.gov) OR your Field Station's Repo Owner to make a request to gain access to the appropriate WERC GitLab folder.

Roles in GitLab



GitLab provides several different roles that can be assigned to users and groups to manage their permissions and access levels. The available GitLab roles include the following:

Role	Access Level	Key Permissions
Guest	Minimal	View project details, leave comments on issue/merge requests, limited interaction

Role	Access Level	Key Permissions
Reporter	Read-Only	View repo, create issues/merge requests, comment on issues/merge requests, pull code without pushing changes
Developer	Write Access	Push code to non-protected branches, create/manage issues/merge requests, review code, comment, and merge, create new branches
Maintainer	Admin Access	All Developer permissions, manage project settings, push to protected branches, create/delete branches and tags
Owner	Full Control	All Maintainer permissions, manage members, transfer projects

WERC Software Release Workflow



WERC Software Checklist



Provisional and Official Release Checklist |
USGS

Check out our [WERC Software Checklist](#) for a break down on the Software Release process.

Provisional Release

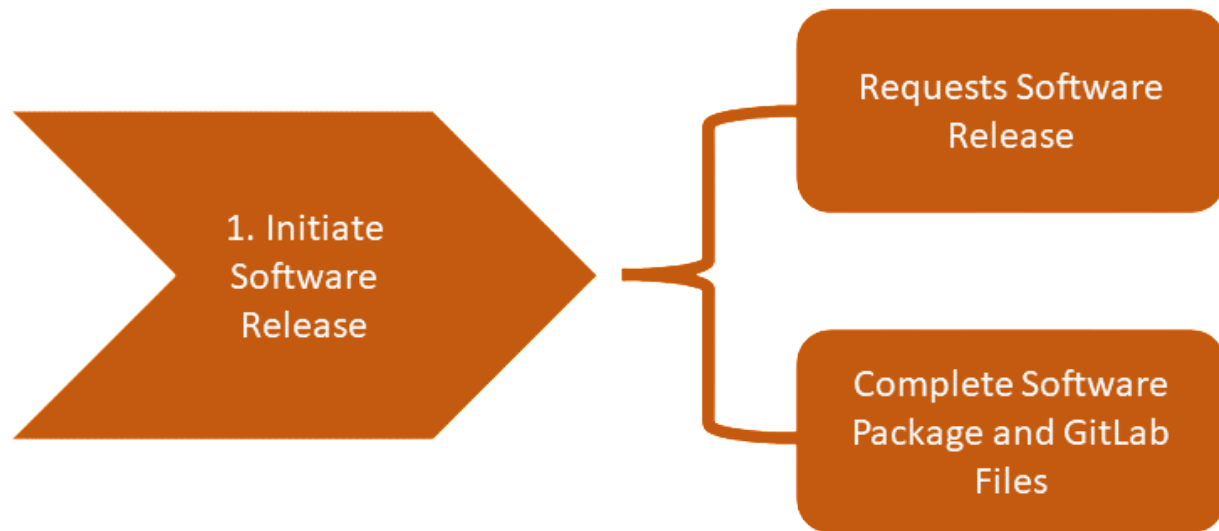


A provisional release is for the entirety of the project. The default branch (i.e. master or main) will be used for the provisional release.

Provisional releases allow the source code to be made available publicly. This may be in the timely interest of best science, to facilitate external domain and/or technical code reviews (in pursuance of a subsequent official release), or for any other reason deemed acceptable by your center director or equivalent or delegated authority.

A provisional release **may not be cited or referenced by other official USGS Information Products** (e.g., manuscripts, data releases). Data produced by a

provisional release may not be used to support any other official USGS Information Product (e.g., manuscript).



1. Initiate Software Release:

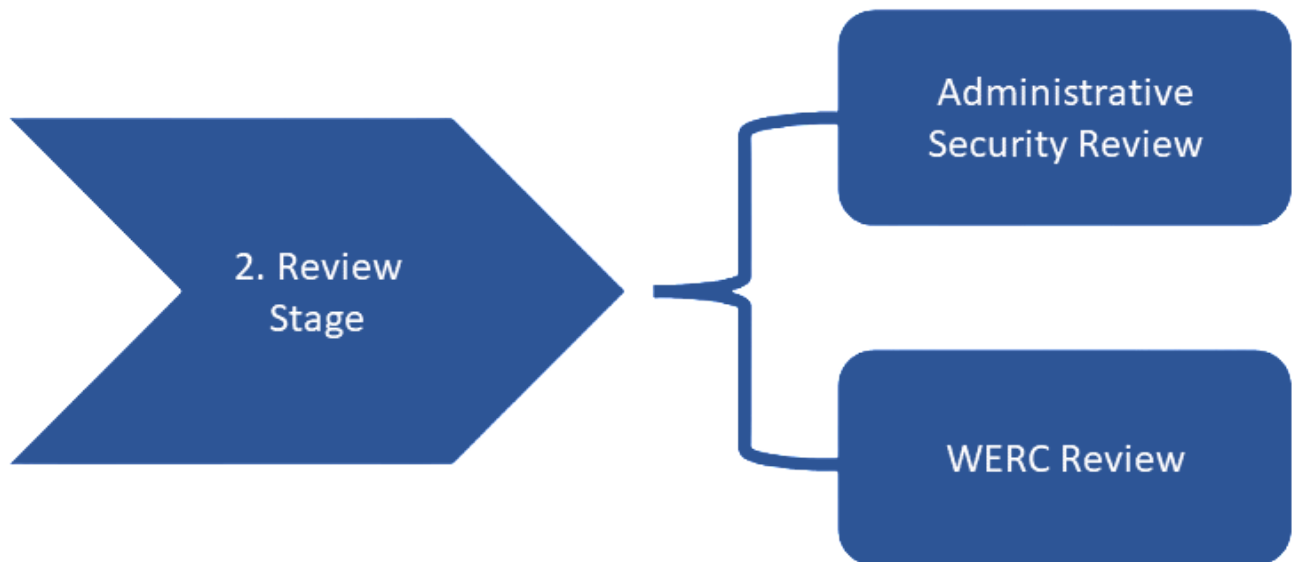
- **Requests Software Release:**

- Email [WERC Data Management group \(gs-b-werc_data_management@usgs.gov\)](mailto:gs-b-werc_data_management@usgs.gov) and CC appropriate [Research Manager](#)
 - Content of email:
 - Title of Release
 - Author List
 - Related IPDS records (paper, data releases)
 - State why you are releasing provisional
 - Will this be released officially in the future?
 - If so, IPDS and DOI create

- **Complete Software Package and GitLab Files:**

- Develops software package on their local computer or Repo

- Author sets up repository with files (See [WERC Repository Template](#) for templates of each file):
 - **README.md:** This will act as the overview page of your software release that relays all the information needed to understand how to use your software. This file should include introductory information about the project (e.g., title, description, use-case, recommended citation, etc.). It should include links to additional detailed information such as the disclaimer statement, license information, documentation, user guide, related manuscript(s), etc...
 - **LICENSE.md:** All software developed by the USGS should by default use an open-source license
 - **code.json:** All USGS software repositories should contain a code.json metadata file and its root. You can create one easily using this template
 - **DISCLAIMER.md:** The appropriate provisional disclaimer statement
 - **.gitignore:** This file is used in a repo to ignore files/directories that we do not want git to push into our remote repository on GitLab. This is not a required file but is useful when pushing/pulling files between RStudio and GitLab. This helps ensure file paths are not included
 - **Your code file(s):** Make sure code is clean. All unnecessary or commented code is removed, proper code syntax is used, lengthy code is well documented with section headers and comments, R functions are documented, and R packages are properly documented



2. Review Stage:

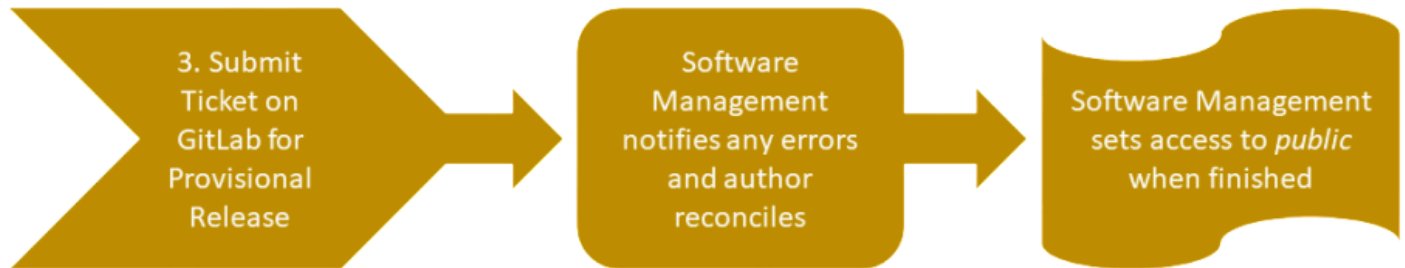
- **Administrative Security Review with Christina Flora:**

- Ensure review inspects each commit in project history
- Ensures that personally identifiable information (PII), absolute file system paths, internal server host names or IP addresses, usernames/passwords, and other personal, private, or otherwise sensitive information is not included with the software
- Email [Christina Flora \(cflora@usgs.gov\)](mailto:cflora@usgs.gov) to start the Administrative Security Review Process
 - An *Issue* will be created in repo
 - [Christina](#) will follow up with feedback on review and close issue

- **WERC Review:**

- Email [Data Manager](#) your repo is ready for review
- Reviews overall repository and README file
 - Ensures files are correctly uploaded in Repo

- README.md is filled out
- All required files are uploaded and filled out correctly



3. Submit Ticket on GitLab for Provisional Release:

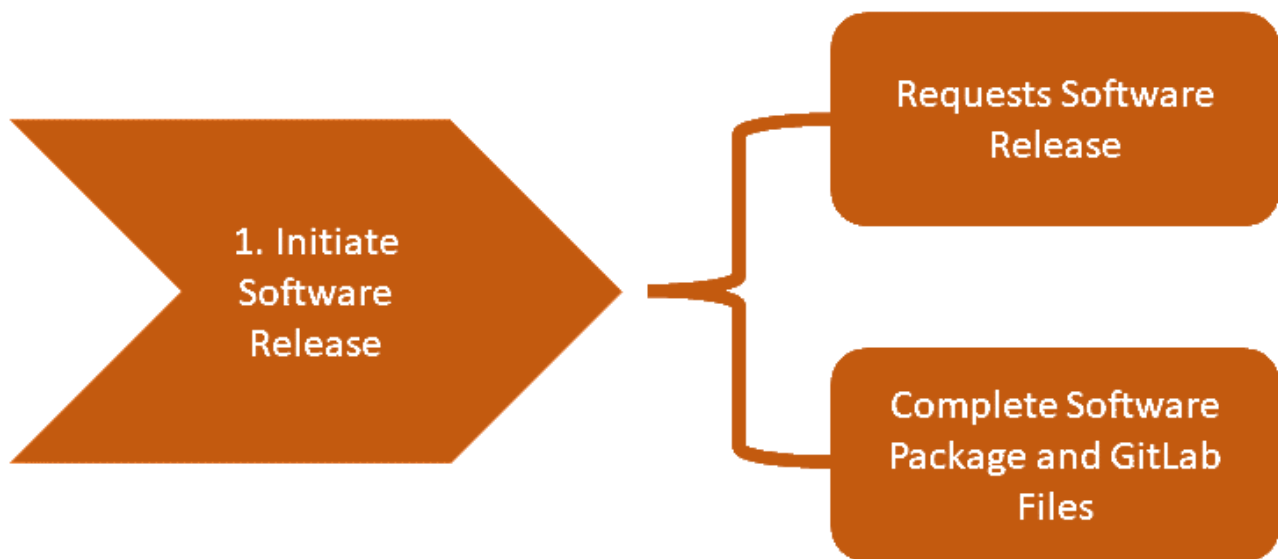
- [Create a ticket](#) with **USGS Software Management** for a GitLab Provisional Release and fill out the template
 - Approving Official: @mdekar
- **USGS Software Management** notifies author of any technical/validation problems
 - Author reconciles
- **USGS Software Management** sets access to *public* when finished
 - Provisional is ready to be shared as needed
 - Remember, a citation cannot be used at this time
 - DOI is not active

Official Release



Official releases must reference immutable Git tags. As you work towards an official release, you can use a release candidate branch to facilitate this process. The release candidate branch should have the same name as the eventual release tag. Upon completing the official release process, all other branches, tags, refs, etc... in the associated GitLab project are automatically released as provisional products as well.

The primary distinguisher between a provisional release and an official release are the contents of the `DISCLAIMER.md` file, the `code.json` status property, and the ability to cite/reference the released product.



1. Initiate Software Release:

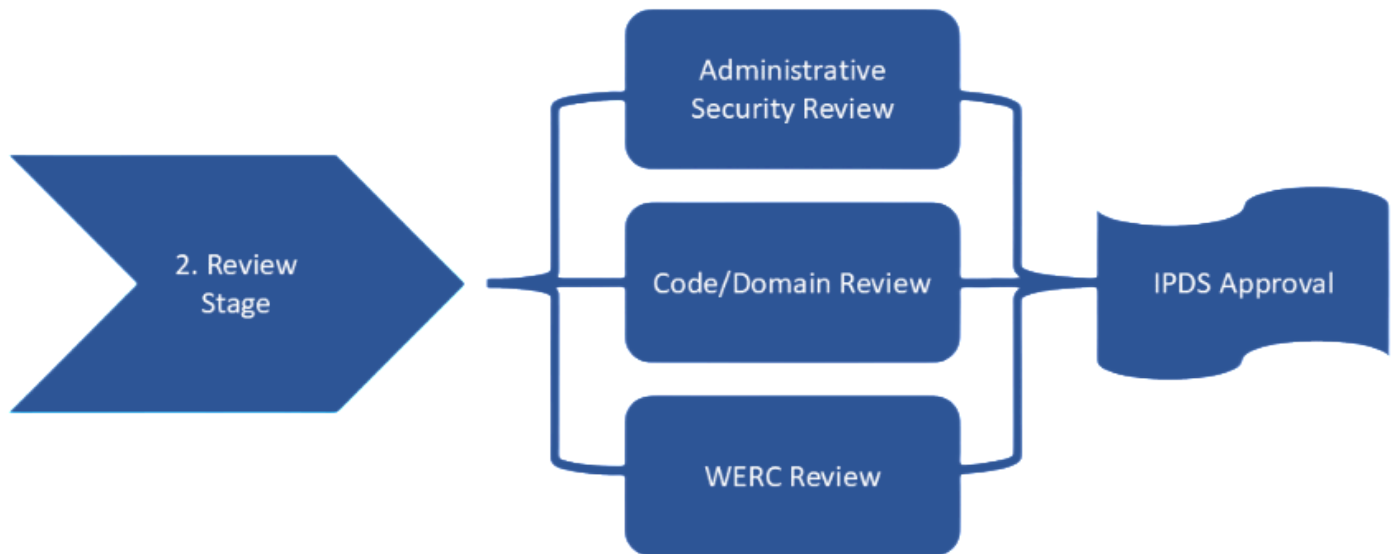
- **Requests Software Release**

- Email [WERC Data Management group \(gs-b-werc_data_management@usgs.gov\)](mailto:gs-b-werc_data_management@usgs.gov) and CC appropriate [Research Manager](#)
 - Content of email:
 - Title of Release
 - Author List
 - Related IPDS records (paper, data releases)
 - Suggested Reviewers for Technical Code and Domain Review
 - [Research Manager](#) approves Reviewer(s)
 - [Data Manager](#) sets up IPDS record and reserves DOI

- **Complete Software Package and GitLab Files:**

- Develops software package on their local computer or Repo

- Author sets up repository with files (See [WERC Repository Template](#) for templates of each file):
 - **README.md:** This will act as the overview page of your software release that relays all the information needed to understand how to use your software. This file should include introductory information about the project (e.g., title, description, use-case, recommended citation, etc.). It should include links to additional detailed information such as the disclaimer statement, license information, documentation, user guide, related manuscript(s), etc...
 - **LICENSE.md:** All software developed by the USGS should by default use an open-source license
 - **code.json:** All USGS software repositories should contain a code.json metadata file and its root. You can create one easily using this template
 - **DISCLAIMER.md:** The appropriate approved disclaimer statement
 - **.gitignore:** This file is used in a repo to ignore files/directories that we do not want git to push into our remote repository on GitLab. This is not a required file but is useful when pushing/pulling files between RStudio and GitLab. This helps ensure file paths are not included
 - **Your code file(s):** Make sure code is clean. All unnecessary or commented code is removed, proper code syntax is used, lengthy code is well documented with section headers and comments, R functions are documented, and R packages are properly documented



2. Review Stage:

- Can have 1 **Reviewer** for Code and Domain if qualified
 - [Templates for each Review](#) for the *Issues* section
- Send email to **Data Manager** once Reviews are finished

• **Administrative Security Review** with **Christina Flora**:

- Guidelines:
 - Ensure review inspects each commit in project history
 - Ensures that personally identifiable information (PII), absolute file system paths, internal server host names or IP addresses, usernames/passwords, and other personal, private, or otherwise sensitive information is not included with the software
- Email **Christina Flora** (cflora@usgs.gov) to start the Administrative Security Review Process
 - An *Issue* will be created in repo
 - **Christina** will follow up with feedback on review and close issue

- **Technical Code Review:**

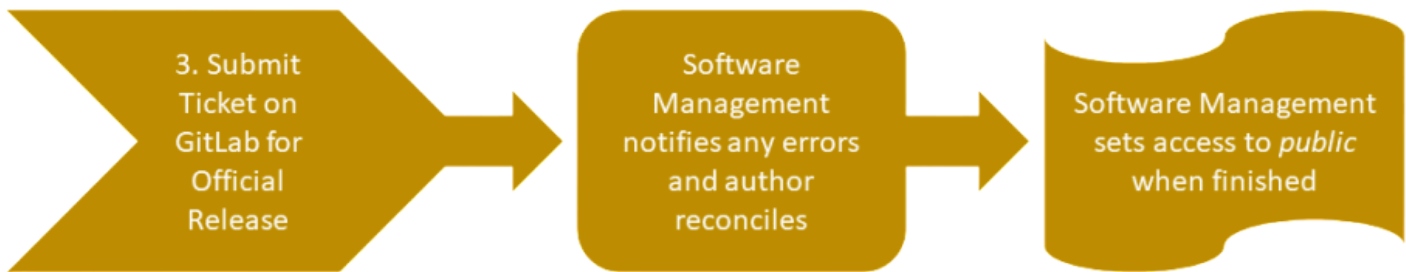
- Guidelines:
 - Can be same person who does the Domain Review *if applicable*
 - Should be completed by somebody with adequate understanding of the programming languages used in the project
- Collaborate with **Reviewer** through preferred method:
 - Author and **Reviewer** share code directly (e.g., email)
 - **Reviewer** sends "Review Document" to Author (e.g., email or Word DOC)
 - Author reconciles code and documentation as needed
 - Document the reconciliation (e.g., email or Word DOC)
 - Facilitated/reconciled via GitLab Issues in WERC Repo
 - Discuss with Field Station Owner to add Reviewer to repo
 - Author assigns Reviewer as a "Developer" of Repo
Manage -> Members -> Invite members -> EMAIL
 - **Reviewer** checks/comments in GitLab Issues
 - Author reconciles code and closes Issue
 - Email documentation to **Data Manager**
 - For GitLab Issues, Data Manager will copy a PDF of the Issue

- **Domain Review:**

- Guidelines:
 - Can be same person who does the Code Review *if applicable*
 - Should be completed by somebody with adequate understanding of the scientific subject matter covered in the released product

- Solid documentation and testing suites can aid this review process
- Collaborate with **Reviewer** through preferred method:
 - Author and **Reviewer** share code directly (e.g., email)
 - **Reviewer** sends "Review Document" to Author (e.g., email or Word DOC)
 - Author reconciles code and documentation as needed
 - Document the reconciliation (e.g., email or Word DOC)
 - Facilitated/reconciled via GitLab Issues in WERC Repo
 - Discuss with Field Station Owner to add Reviewer to repo
 - Author assigns Reviewer as a "Developer" of Repo
 - Manage -> Members -> Invite members -> EMAIL
 - **Reviewer** checks/comments in GitLab Issues
 - Author reconciles code and closes Issue
 - Email documentation to **Data Manager**
 - For GitLab Issues, Data Manager will copy a PDF of the Issue
- **WERC Review with Data Manager:**
 - Email **Data Manager** your repo is ready for review (after Code, Domain, and Security)
 - Reviews overall repository and README file
 - Ensures README is filled out correctly
 - DOI is listed
 - Citations formatted correctly
 - All required files are uploaded and filled out correctly
- **IPDS Approval:**

- **Research Manager** approves Software Release in IPDS
- **Center Director** approves Software Release in IPDS
- **Data Manager** notifies author approval from Center Director



3. **Submit Ticket on GitLab for Official Release:**

- [Create a ticket](#) with **USGS Software Management** for a **GitLab Official Release** and fill out the template
 - **Approving Official:** @mdekar
 - Be sure to create a *Versioned* branch for your ticket
- **USGS Software Management** notifies author of any technical/validation problems
 - Author reconciles
- **USGS Software Management** sets access to *public* when finished
 - Email **Data Manager** to initiate final steps



4. Repo Review and Publication:

- **Tag and Release Created:**

- Create version tag
- Delete *Versioned* branch
- Create *Release* branch and ensure tag (labeled as the version number) is set up

- **Published!**

- **Data Manager** sets DOI to go live
- **Data Manager** disseminates IPDS Record
- **Data Manager** emails author Software Release is published with citation

Revisions

Revisions are classified into 3 levels:

- **Major (e.g., v2.0.0)**

- Rewrites of major sections of code, significant reorganization of code, changes to equations (excluding corrections), multiple new functionality (i.e., features), and changes that prevent backward compatibility (e.g., addition or removal of features, changes to database structures, or application programming interface [API] commands).
- Review: Code, Subject, Security - required
- New IPDS and DOI records

- **Minor (e.g., v1.1.0)**

- Modifications intended to address minor changes that will not significantly impact the use of the software, will introduce limited new functionality, will not include extensive enhancements/extensions to existing features, may include revisions to documentation, and address security bug fixes such as deprecated features and necessary updates to dependent library packages. These changes should not affect compatibility of using functions from previous code (e.g., unchanged code still behaves the same and structure of code organization not overhauled).
- Review: Security - required
- Code & Subject - suggested after 2-3 minor revisions
- New IPDS record

- **Patch (e.g., v1.0.1)**

- Modifications intended to address bug fixes, corrections to documentation, and security bug fixes such as deprecated features and necessary updates to dependent library packages. These changes should include minor changes to address broken functionality or security risks.
- Review: Security - required
- Code & Subject - suggested after several patch revisions

Q&A

Questions and Answers

- **I followed the instructions to setting up R with GitLab and I cannot push my files. What could be the issue?**

- First check what your *Role* is in your GitLab repo. If you don't have *Maintainer* or *Owner* permissions, you cannot push files locally (on R) to GitLab.
- If you have a Mac, this may be due to permissions. I would recommend asking [CDI Software](#) for more help.

- **If I am updating/revising a shiny application, can I use a different project file on R?**
 - In order to keep the same Rconnect link, the project would need to be published from the same project file.
-